

A Drug-Drug Interaction of AINUOVIRINE with Fluconazole in Healthy Adults and Dose Optimization Informed by Pharmacokinetics Modelling

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Purpose: The study aimed to assess the drug-drug interaction (DDI) and safety of ainuovirine (a new-generation non-nucleoside reverse transcriptase inhibitor, primarily metabolized by CYP2C19) coadministered with fluconazole, a strong CYP2C19 inhibitor, in healthy adults.

Methods: This was a multiple-dose, open-label, two-period, parallel, sequential study. In group A, 18 participants received ainuovirine (150 mg, q.d.) from day 1–7 (Period 1), followed by coadministration with fluconazole (200 mg, q.d.) from day 8–14 (Period 2). In group B, 18 participants received fluconazole (200 mg, q.d.) from day 1–7 (Period 1), followed by coadministration with ainuovirine (150 mg, q.d.) from day 8–14 (Period 2). A physiologically based pharmacokinetic (PBPK) model of ainuovirine predicted its DDI with fluconazole, incorporating population PK analysis.

Results: In group A, when coadministered with fluconazole, geometric means of ainuovirine $C_{\max,ss}$, $AUC_{0-24,ss}$ increased up to 233.0% and 349.6%, respectively, versus ainuovirine alone. In group B, there were no apparent effects of ainuovirine on $C_{\max,ss}$, $AUC_{0-24,ss}$ and $T_{\max,ss}$ for fluconazole. No death or grade ≥ 3 serious adverse events occurred. The predictive performance of the PBPK model for the interaction scenario is excellent, because the ratios of predicted-to-observed $C_{\max,ss}$ and $AUC_{0-\infty,ss}$ are very close to 1.0, which provides high confidence in the model's simulations for dose optimization.

Conclusion: Coadministration with fluconazole significantly alters the steady-state PK of ainuovirine, but not conversely. The DDI PBPK model supports that dose reduction of ainuovirine is warranted for coadministration with fluconazole.

Keywords: CYP2C19, PBPK modelling, clinical trial, clinical relevance

Introduction

Human immunodeficiency virus (HIV) is a transmissible infection that attacks the human immune system by destroying CD4 T lymphocyte cells. The global number of people living with HIV (PLWH) had reached 40.8 million by the end of 2024. HIV-1 is the most dominant strain in China. The PLWH face an increased risk and persistent burden of opportunistic fungal infections. Fungal infections are major and frequent life-threatening illnesses among the PLWH, especially in developing countries, owing to late diagnosis and limited antiretroviral therapy (ART) access.¹ The diagnosis, treatment, and precaution of HIV-related fungal infections remain challenging in China, owing to poor

adherence and/or ART resistance. These opportunistic fungal infections include candidiasis, pneumocystosis, cryptococcosis, histoplasmosis and talaromycosis.^{1,2} It was estimated that 47% of acquired immunodeficiency syndrome (AIDS) deaths are caused by fungal infections annually.³

Non-nucleoside reverse transcriptase inhibitors (NNRTIs), as a class of antiretroviral drugs, have been used to treat HIV-1 infection by targeting the HIV-1 reverse transcriptase and inhibiting the enzyme's activity.⁴ FDA-approved NNRTIs for treating HIV-1 include nevirapine, delavirdine, efavirenz, etravirine, rilpivirine and doravirine.⁵ NNRTIs, with their distinctive antiretroviral activity and high specificity, have been used in both first-line and alternative regimens, often in combination with two nucleoside reverse transcriptase inhibitors (NRTIs) for treatment-naïve adults with HIV-1. However, NNRTIs still face major challenges related to their relatively low genetic barrier to resistance.⁶ Among these, efavirenz, as a first-generation NNRTI, combined with tenofovir disoproxil fumarate and lamivudine (or emtricitabine) is a WHO-recommended first-line regimen for PLWH initiating ART. However, efavirenz has limited oral bioavailability owing to its poor water solubility and extensive first-pass metabolism in the liver.^{7,8} Efavirenz is associated with central nervous system toxicities, a low genetic barrier to resistance and increases drug-drug interactions (DDI) due to its role as a moderate inducer of CYP3A4.^{9,10} AINUOVIRINE is a new-generation NNRTI for treatment-naïve adults with HIV-1, which is predominantly metabolized by CYP2C19 and does not undergo significant metabolism by CYP2C9, CYP3A4, CYP1A2, CYP2B6, CYP2C8, or CYP2D6.^{11,12} AINUOVIRINE is slowly metabolized by CYP2C19 and exhibits CYP2C19 induction potential.¹¹ CYP2C19 gene polymorphism has relatively little effect on the clearance of AINUOVIRINE.¹¹ AINUOVIRINE has a mild induction effect on CYP3A4.¹³ AINUOVIRINE is not a substrate for the P-glycoprotein (P-gp) transporter, therefore its blood concentration is not affected by P-gp.¹¹ AINUOVIRINE (75 mg, Tablets) has been approved by the Chinese National Medical Products Administration since June 2021 for use in combination with other ART regimen in HIV-1-infected treatment-experienced adults. AINUOVIRINE has shown its effectiveness and safety in a previous Phase III clinical trial.¹² AINUOVIRINE (150 mg q.d.) shows comparable efficacy to efavirenz (600 mg q.d.) when combined with lamivudine and tenofovir disoproxil fumarate. Furthermore, AINUOVIRINE is associated with less hepatotoxicity and fewer neuropsychiatric symptoms than efavirenz.¹²

Fluconazole is an antifungal drug used to treat patients with candidiasis and cryptococcal disease.^{14,15} Fluconazole has high oral bioavailability and is well absorbed.¹⁶ Fluconazole is poorly metabolized, primarily eliminated by the kidneys, and has a plasma elimination half-life of approximately 30 h in healthy subjects.¹⁷ As a strong cytochrome P450 2C19 (CYP2C19) inhibitor, fluconazole significantly affects the pharmacokinetics (PK) of CYP2C19 substrates.^{18,19} Fluconazole has been found to potently inhibit CYP2C19 both in vitro¹⁸ and in vivo.²⁰ Dose reduction is required for drugs metabolized by CYP2C19, such as nelfinavir,²¹ omeprazole,²⁰ and diazepam,²² when coadministered with fluconazole. Fluconazole is also a moderate inhibitor of CYP2C9 and CYP3A4.²³ Thus, it is important to understand the potential drug-drug interactions (DDIs) of fluconazole.

DDI between AINUOVIRINE and strong CYP2C19 inhibitors should be considered for PLWH with fungal infections. The aim of the study was to assess the DDI profile and safety of AINUOVIRINE when coadministered with fluconazole in healthy adults. A physiologically-based pharmacokinetics (PBPK) model was developed to predict the DDI between AINUOVIRINE and fluconazole.

Methods

Study Design

There was an inpatient study at the Guangzhou Eighth People's Hospital (Guangzhou, China) from April 2023 to August 2023. The detailed inclusion and exclusion criteria of the study population are described in the supplementary material. This was an open-label, multiple-dose, parallel, sequential study consisting of a 14-day screening period followed by two treatment periods. According to the ICH guidelines for Drug Interaction studies M12, in section 3.2.1.5, parallel, two-arm studies can be appropriate when a crossover study design is not feasible, such as when one of the drugs has a long half-life. Parallel study designs instead of cross-over designs have been suggested as options for drugs with long half-life in order to avoid carry-over. Fluconazole has a relatively long elimination half-life ($t_{1/2} \approx 32$ h)²⁴

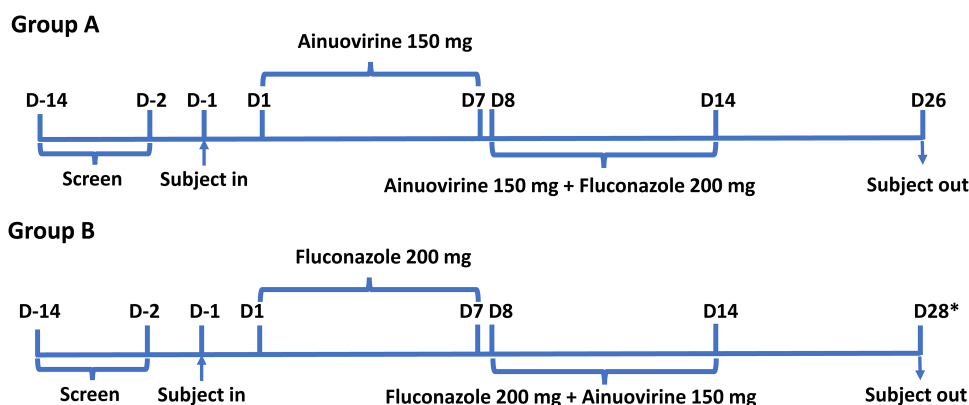


Figure 1 Study design and dosing, for group A and group B. In group A, participants received ainuovirine (150 mg, q.d.) from day 1–7, followed by coadministration with fluconazole (200 mg, q.d.) from day 8–14. In group B, participants received fluconazole (200 mg, q.d.) from day 1–7, followed by coadministration with ainuovirine (150 mg, q.d.) from day 8–14. * If plasma concentration in 336 h (D28) was below the limit of quantification (BQL) for the two subjects in the pilot study, then the date of subject out would be updated to D26 for the other subjects enrolled in Group B.

and its plasma half-life is dose independent.²⁵ The use of parallel study design in other DDI studies of fluconazole has also been reported.^{26–28} Thus, parallel study design is preferred over crossover study design in the DDI study.

The 150 mg dose of ainuovirine was selected based on its efficacy and safety shown in a previous phase III trial.¹² A clinically relevant 200 mg dose of fluconazole was used for the treatment of oropharyngeal candidiasis.¹⁴ The plan was to include 36 subjects. As shown in Figure 1, in group A, 18 healthy adults received oral ainuovirine (150 mg, q.d.; dosage form: tablet, manufacturer: Jiangsu Aidi Pharmaceuticals Co., Ltd, Jiangsu, China) from day 1–7 (Period 1), followed by oral coadministration with fluconazole (200 mg, q.d.; dosage form: capsule; manufacturer, Pfizer Inc., (New York, NY) from day 8–14 (Period 2). In group B, 2 subjects were admitted to the study site on day –1 as a pilot study. The healthy adults received oral fluconazole (200 mg, q.d.) from day 1–7 (Period 1), followed by oral coadministration with ainuovirine (150 mg, q.d.) from day 8–14 (Period 2). After the first 2 subjects completed the study, the remaining 16 subjects were enrolled and followed the same procedure. All doses were taken with 240 mL of water after an overnight fast (≥ 10 h, with no fluid intake within the 1 h before and after dosing), followed by 4 hours of post-dose fasting. Subjects received standard diet during the study period. Safety was evaluated approximately 12 days after the last dose.

Safety and Tolerability Assessments

Safety was assessed by adverse events (AEs), laboratory parameters, vital signs, physical examinations and 12-lead ECGs (electrocardiogram), all of which were obtained at prespecified times throughout the study. All adverse events were recorded and characterized in terms of their clinical manifestations, severity, onset and duration, action taken on treatment, subject outcome and relationship to treatment as judged by the investigator. All treatment-emergent AEs (TEAEs) were coded and classified using the Medical Dictionary for Regulatory Activities (MedDRA, version 26) preferred terms and TEAE severity was graded using the National Cancer Institute—Common Terminology Criteria for AEs (NCI-CTCAE, version 5.0).

PK Blood Sampling Schedule and Bioanalysis

At each time point, 3 mL of blood was collected from each subject. In group A, during period 1, blood samples for ainuovirine PK analysis were drawn predose on day 1, 2, 5, 6 and 7, and postdose on day 1 and 7 at 0.5, 1, 2, 3, 4, 8 and 12 h. During period 2, blood samples for both drugs PK analysis were drawn predose on day 8, 9, 12, 13, and 14, and postdose on day 8 at 0.5, 1, 2, 3, 4, 8, and 12 h, and on day 14 at 0.5, 1, 2, 3, 4, 8, 12, 24, 48, 72, 144, 168, 192, 216, and 288 h.

In group B, during period 1, blood samples for fluconazole PK analysis were drawn predose on day 1, 2, 5, 6 and 7, and postdose on day 1 and 7 at 0.5, 1, 2, 3, 4, 8 and 12 h. During period 2, blood samples for both drugs PK analysis were

drawn predose on day 8, 9, 12, 13 and 14, and postdose on day 8 at 0.5, 1, 2, 3, 4, 8, 12 h, and on day 14 at 0.5, 1, 2, 3, 4, 8, 12, 24, 48, 72, 144, 168, 192, 216, 288, 336 h*. [*blood samples were collected from pilot study subjects at 336 h (day 28) to confirm whether the elimination phase was complete. If the plasma concentration at 336 h (day 28) was below the limit of quantification (BLQ) in the pilot study, PK samples would not be collected at this time point for other subjects in group B.] For both groups, plasma samples of ainuovirine and fluconazole were analysed by liquid chromatography-mass spectrometry (LC-MS/MS). An Acquity UPLC I-Class (Waters, UK) was coupled with the electro-spray ionization (ESI) source (Xevo TQS, waters, USA). Chromatographic separations were performed on an Acquity UPLC BEH C18 column at 2.1×50 mm, 1.7 µm (Waters, UK). The samples were eluted at a flow rate of 450 µL/min using a gradient mixture of 0.1% (v/v) formic acid aqueous solution (solvent A) and acetonitrile (solvent B). The gradient program was as follows: 0.0–3.0 min, 20% B; 3.0–4.0 min, linear increase from 20% to 95% B; 4.0–5.0 min, linear decrease from 95% to 20% B, with a column temperature of 40 °C. The injection volume was set at 2 µL. The retention times of ainuovirine and fluconazole were 1.79 min and 0.69 min, respectively. Mass spectrometry was performed using the ESI source in positive ion multiple reaction monitoring mode. Ainuovirine-d5 (internal standard [IS]) (98.9%) was obtained from Nanjing Accelas Pharmaceutical Co., Ltd. (Nanjing, China). Fluconazole-d4 (IS) (95.5%) was obtained from Beijing Manhage Bio-Technology Co., Ltd. (Beijing, China). Quantification was carried out at ion transitions m/z 326.1 → 298.1 for ainuovirine and m/z 331.1 → 299.1 for ainuovirine-d5 (IS), m/z 307 → 219.9 for fluconazole, m/z 311.1 → 223 for fluconazole-d4 (IS). Quantification of ainuovirine was done using a calibration curve, which was constructed by diluting a stock solution of ainuovirine (1005.98 µg/mL) in methanol:water (90:10, v/v) and linear calibration curve ($R^2 = 0.9958$) in the range of 4–1000 ng/mL was obtained. The calibration curve of fluconazole was constructed by diluting a stock solution of fluconazole (1999.95 µg/mL) in methanol:water (90:10, v/v) and linear calibration curve ($R^2 = 0.9983$) in the range of 80–20000 ng/mL was obtained. The lower limit of quantitation for ainuovirine and fluconazole was 4 ng/mL and 80 ng/mL, respectively.

Pharmacokinetic Analyses

PK parameters of ainuovirine were determined from the blood concentration-time profiles both when administered alone (period 1) and with fluconazole (period 2) by noncompartmental analysis using Phoenix[®] WinNonlin[®] version 8.3 software (Certara, NJ, USA). PK parameters of fluconazole were obtained by the same method.

The PK analysis set included all subjects with data for PK variable of C_{max} , AUC, the average concentration (C_{av}), time to reach maximum plasma concentration (T_{max}), plasma oral clearance (CL/F), $T_{1/2z}$, apparent volume of distribution during terminal phase (V_z/F) at steady state, and other PK parameters for ainuovirine and fluconazole. These parameters were descriptively analyzed by treatment group.

To assess the DDI effect, a mixed model was fitted to the log-transformed PK parameters $C_{max,ss}$ and AUC_{ss} of ainuovirine and fluconazole. The model included treatment as a fixed effect and variance within subjects as a random effect. The geometric least squares mean ratio and the 90% confidence interval (CI) of the $C_{max,ss}$ and AUC_{ss} between the two different treatments of each group were determined. It was concluded that ainuovirine had no effect if the 90% CI for the ratios of AUC or C_{max} , both with and without fluconazole, were within 80–125%. According to FDA industrial guidelines on Statistical Approaches to Establishing Bioequivalence, bioequivalence (BE) is established when the calculated CI falls within the BE limit of 80–125% for the ratio of geometric means. Therefore, the interval 80–125% is used as a reference in this study.

Statistical Evaluation

Statistical analyses were conducted using SAS (version 9.4; SAS Software Inc., Cary, NC, USA) unless otherwise specified. Safety assessments included all enrolled subjects who received at least one dose of the study drug. PK assessments were based on all enrolled subjects who received at least one dose of the study drug and had at least one valid PK concentration measurement available.

PBPK Model Development

In this study, population PBPK modelling and simulation was performed using PK-SIM[®] software (version 11.3, Open system Pharmacology). A model for ainoovirine was developed through the integration of physicochemical parameters, in vitro data, and in silico predictors to predict the absorption, distribution, metabolism, and excretion processes. Virtual populations comprising 1000 individuals were generated based on the demographic characteristics of the study populations obtained from each clinical study. For distribution, a preclinical mouse intravenous PBPK model was initially established and validated based on fraction unbound in plasma and blood-to-plasma concentration ratio data. After a single intravenous administration of 1 mg/kg of ainoovirine to a mouse, the measured volume of distribution was 3.4 L/kg, closely matching the volume of distribution (ie., 3.37 L/kg) calculated by the Poulin and Theil method. Thus, the steady-state volume of distribution (V_{ss}) of ainoovirine was predicted using the Poulin and Theil method, consistent with the value observed in the mouse model. Considering that ainoovirine was predominantly eliminated via the liver, hepatic clearance was implemented and estimated during the parameter optimization step. The relative tissue-specific expression of the enzyme predominantly involved in the metabolism of ainoovirine (CYP2C19) was considered. CYP2C19 was the only pathway considered for the metabolism of ainoovirine in this study, owing to the slow metabolism of ainoovirine by CYP2C19 and the lack of significant metabolism by other major CYP enzymes, including CYP1A2, CYP2B6, CYP2C8, CYP2C9, CYP2D6, and CYP3A4.¹¹ Based on the clinical trial report of healthy volunteers administered with the ¹⁴C-labeled ainoovirine, the urinary recovery of unchanged parent drug accounted for only 0.7% of the total administered dose. As the current model was developed for the parent drug of ainoovirine and did not involve metabolites, the drug in the model was assumed to be eliminated entirely via hepatic clearance. Thus, metabolic enzyme activity was described as first order process and fm (fraction metabolized) was set to 1. The specific clearance ($CL_{spec}/Enzyme$) of ainoovirine was optimized from the experimentally obtained concentration-time profiles in the clinical study. Clinical data in participants with HIV (ADYY-ACC007-103, unpublished) showed dose- and time-dependent reduction in drug exposure upon multiple dosing, consistent with auto-induction of CYP2C19-mediated metabolism. Thus, CYP2C19 induction was considered for ainoovirine. The EC_{50} and E_{max} value of ainoovirine were optimized from the plasma concentration-time profiles in the clinical study. The dissolution kinetics of ainoovirine tablet were described by a Weibull function.

A previously reported PBPK model of fluconazole was integrated with the model of ainoovirine to predict their potential DDI. The PBPK model for fluconazole in adults was previously developed using PK-SIM[®] by Watt et al²⁹ and refined by Cristofolletti et al³⁰. In this study, the K_i (inhibitory constant) value of fluconazole (ie., 3.50 mM) for CYP2C19-dependent inhibition was referenced from the GastroPlus[®] database and the K_i value was used for the establishment of fluconazole and ainoovirine DDI prediction in PK-SIM.

PBPK Model Verification

Simulations of the first dose and multiple doses (75 mg and 150 mg) of ainoovirine were performed. Each simulation result was compared against the experimental data from unpublished studies (ADYY-ACC007-103 and ADYY-ACC007-401), with 90% prediction intervals described. Current visual assessment of ideal model performance is based on whether the observed plasma concentration–time datasets fall within the 90% prediction intervals. Current quantitative assessment of good model prediction performance is based on whether the ratios of predicted to observed AUC and C_{max} fall within 0.5- to 2.0-fold.

Results

Subjects

Overall, 142 healthy adults were screened, of whom 36 were enrolled, with 18 each in groups A and B. All 36 subjects (100%) received all treatments and completed the study, and were included in the safety and PK analysis sets. Group A included 12 subjects from Han ethnicity, and 6 subjects from other ethnicities, including three Zhuang, one Yao, one Miao, and one Yi. Group B included 17 subjects from Han ethnicity and 1 subject from another ethnicities, ie., Miao. There were no substantial differences between the two groups in baseline characteristics, although group B included more healthy Chinese Han subjects than group A (Table 1).

Table 1 Demographics and Baseline Characteristics of Study Subjects

Characteristic	Mean $\pm$ SD		P value
	Group A (n=18)	Group B (n=18)	
Age (y)	27.6 $\pm$ 4.4	30.4 $\pm$ 4.9	0.08
Height (cm)	164.8 $\pm$ 6.8	165.6 $\pm$ 6.9	0.73
Body weight (kg)	60.7 $\pm$ 7.0	60.7 $\pm$ 7.1	>0.99
Body mass index (kg/m ²)	22.3 $\pm$ 1.8	22.1 $\pm$ 1.9	0.75
No. (%) of males	15 (83.3)	15 (83.3)	>0.99
No. (%) of females	3 (16.7)	3 (16.7)	>0.99
No. (%) of Han ethnicity	12 (66.7)	17 (94.4)	0.02
No. (%) of Other ethnicities	6 (33.3)	1 (5.6)	0.02
No. (%) of married	3 (16.7)	2 (11.1)	0.64
No. (%) of unmarried	15 (83.3)	16 (88.9)	0.64
ALT (IU/L)	16.5 $\pm$ 8.0	19.4 $\pm$ 9.2	0.32
AST (IU/L)	16.7 $\pm$ 3.7	18.9 $\pm$ 4.2	0.10
CRE (μ mol/L)	73.3 $\pm$ 12.6	75.2 $\pm$ 11.5	0.64
Uric acid (μ mol/L)	349.0 $\pm$ 77.7	362.3 $\pm$ 64.1	0.58
Urea (mmol/L)	3.9 $\pm$ 1.2	3.5 $\pm$ 0.8	0.21

Abbreviations: SD, standard deviation; ALT, Alanine aminotransferase; AST, Aspartate aminotransferase; CRE, Creatinine.

Effect of Fluconazole on AINUOVIRINE Pharmacokinetics

The $C_{\max}$ and AUC_{0-24} of aINUOVIRINE increased when it was coadministered with fluconazole, compared to when aINUOVIRINE was administered alone (Figure 2a and b). The inter-individual variability (%CV) for aINUOVIRINE $AUC_{0-24,ss}$ was reduced from 39% when given alone to 15% when co-administered with fluconazole. The results also showed that the $V_{Z_{ss}}/F$ (L) and CL_{ss}/F (L/h) of aINUOVIRINE were decreased when co-administered with fluconazole (group A in Table 2). Moreover, the $t_{1/2}$ of aINUOVIRINE was longer when coadministered with fluconazole (group A in Table 2). The 90% CIs for the geometric mean ratios of $C_{\max}$ and AUC_{0-24} for aINUOVIRINE (with vs. without fluconazole) exceeded the prespecified interval of 80–125% (group A in Table 3).

Effect of AINUOVIRINE on Fluconazole Pharmacokinetics

Mean concentration-time profiles after multiple doses of fluconazole appeared similar, whether administered with or without aINUOVIRINE (Figure 2c and d). No difference in $C_{\max}$ and AUC_{0-24} was observed between the reference (fluconazole) and the test (fluconazole + aINUOVIRINE) groups (group B in Table 2). The 90% CIs for the geometric mean ratios of $C_{\max}$ and AUC_{0-24} for fluconazole (with vs. without aINUOVIRINE) were within the prespecified interval of 80–125% (group B in Table 3).

Safety and Tolerability

In group A, AEs occurred in 83.3% (15/18) of subjects, including fifteen subjects (83.3%) following the administration of aINUOVIRINE and fourteen subjects (77.8%) following the administration of aINUOVIRINE with fluconazole (Table 4). The following AEs were mainly observed: diastolic blood pressure decreased, abnormal ECG T waves, and blood triglycerides increased (Table S1). The incidence of TEAEs was higher when aINUOVIRINE was coadministered with fluconazole compared to when aINUOVIRINE was given alone. All of the AEs were rated as mild in severity (grade 1) by the investigator. There were no $\geq$ grade 3 TEAEs, no severe adverse events and no deaths during the study. No subjects withdrew due to AEs during the study.

In group B, AEs occurred in 94.4% (17/18) of subjects, including twelve subjects (66.7%) following the administration of fluconazole and sixteen subjects (88.9%) following the administration of fluconazole with aINUOVIRINE (Table 5). The following AEs were mainly observed: diastolic blood pressure increased, systolic blood pressure decreased, heart rate decreased, and blood triglycerides increased (Table S2). The incidence of TEAEs was higher

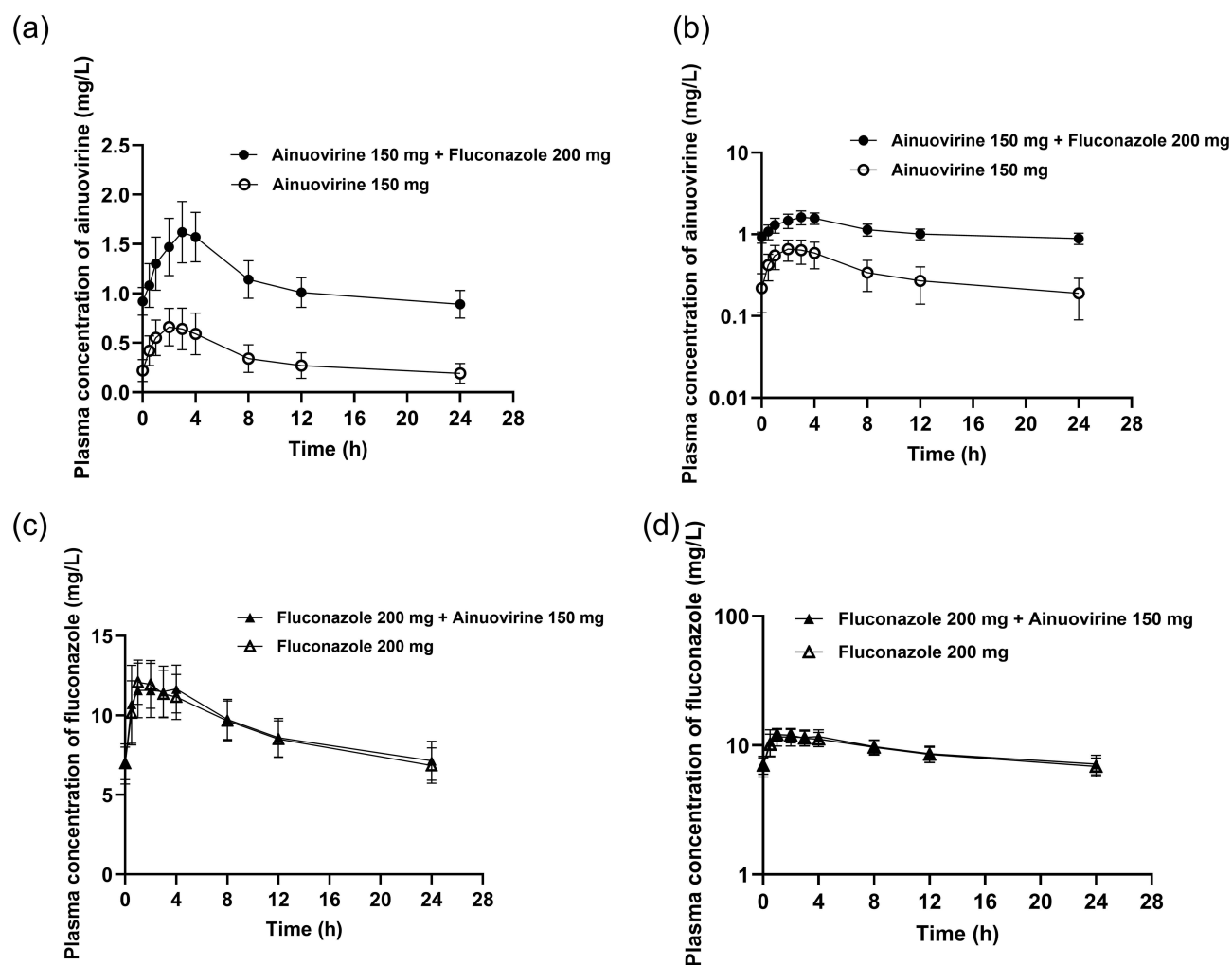


Figure 2 Arithmetic mean (SD) plasma concentration-time curves. (a) linear and (b) semi-logarithmic profiles of ainuovirine with or without coadministration of fluconazole; (c) linear and (d) semi-logarithmic profiles of fluconazole with or without coadministration of ainuovirine.

Abbreviation: SD, standard deviation.

when fluconazole was coadministered with ainuovirine compared to when fluconazole was given alone. The majority of AEs were rated as mild (grade 1) and only three events were rated as moderate in severity (grade 2) by the investigator. There were no $\geq$ grade 3 TEAEs, no severe adverse events and no deaths during the study. No subjects withdrew due to AEs during the study.

Table 2 Plasma PK Parameters for Multiple Doses of Ainuovirine and Fluconazole (PK Analysis Set)

PK Parameters	Ainuovirine (Group A) N = 18		Fluconazole (group B) N = 18	
	Ainuovirine Alone Period 1 on Day 7	Ainuovirine + Fluconazole Period 2 on Day 14	Fluconazole ALONE Period 1 on Day 7	Fluconazole + Ainuovirine Period 2 on Day 14
AUC _{ex} (%)	35.8 ± 11.3	0.5 ± 0.4	58.7 ± 6.7	1.0 ± 0.4
AUC _{0-∞,ss} (h mg/L)	13.11 ± 6.6	67.78 ± 16.10	533.62 ± 180.07	524.33 ± 128.13
AUC _{0-24,ss} (h mg/L)	8.02 ± 3.16	26.70 ± 4.02	213.83 ± 29.11	217.95 ± 30.81
C _{av,ss} (mg/L)	0.33 ± 0.13	1.11 ± 0.17	8.91 ± 1.21	9.08 ± 1.28
C _{max,ss} (mg/L)	0.73 ± 0.21	1.67 ± 0.3	12.68 ± 1.48	12.65 ± 1.78

(Continued)

Table 2 (Continued).

PK Parameters	Ainuovirine (Group A) N = 18		Fluconazole (group B) N = 18	
	Ainuovirine Alone Period 1 on Day 7	Ainuovirine + Fluconazole Period 2 on Day 14	Fluconazole ALONE Period 1 on Day 7	Fluconazole + Ainuovirine Period 2 on Day 14
$C_{min,ss}$ (mg/L)	0.19 ± 0.10	0.91 ± 0.14	6.75 ± 1.08	6.94 ± 1.31
CL_{ss}/F (L/h)	21.66 ± 8.71	5.73 ± 0.81	0.95 ± 0.13	0.94 ± 0.14
DF (%)	173.7 ± 45.9	67.2 ± 14.5	67.4 ± 14.2	63.7 ± 15.9
Rac(AUC)	1.1 ± 0.2	1.8 ± 0.3	2.4 ± 0.33	1.0 ± 0.04
Rac(C_{max})	1.2 ± 0.3	1.6 ± 0.2	2.3 ± 0.4	1.0 ± 0.1
$T_{max,ss}$ (h)	2.3 ± 1.1	2.8 ± 0.9	1.4 ± 0.7	2.0 ± 1.5
$V_{z,ss}/F$ (L)	518.56 ± 246.57	227.12 ± 107.73	42.86 ± 10.19	39.63 ± 5.16
$t_{1/2,ss}$ (h)	17.3 ± 6.2	27.3 ± 10.7	31.7 ± 9.6	29.7 ± 4.6
$\lambda_{z,ss}$ (1/h)	0.05 ± 0.02	0.03 ± 0.01	0.02 ± 0.01	0.02 ± 0.004

Notes: Data are presented as Mean ± SD (standard deviation). N = number of subjects that received treatment in the study period.

Abbreviations: $AUC_{0-\infty}$ (%), percentage of the area under the concentration-time curve that has been derived after extrapolation; $AUC_{0-24,ss}$, area under the plasma concentration-time curve from time 0 to 24 h; $C_{av,ss}$, average steady-state plasma concentration; $C_{max,ss}$, maximum steady-state plasma concentration; $C_{min,ss}$, minimum steady-state plasma concentration, PK pharmacokinetic; $T_{max,ss}$, time to reach maximum concentration at steady-state; CL_{ss}/F , plasma steady-state oral clearance; DF fluctuation coefficient; DF = $100\% \times (C_{max,ss} - C_{min,ss})/C_{av,ss}$; Rac, accumulation index; RA (AUC) = multiple-dose AUC_{0-t} / single-dose AUC_{0-t} ; RA (C_{max}) = $C_{max,ss}/C_{max}$; $V_{z,ss}/F$, apparent volume of distribution at steady state; $t_{1/2,ss}$, elimination half-life; $\lambda_{z,ss}$, elimination rate constant of drug.

Table 3 Summary of Log Transformed Statistical Analysis of Drug-Drug Interactions

PK Parameters	Ainuovirine (Group A)		GM Ratio (%)	90% CI (%)	Fluconazole (Group B) (GM)		GM Ratio (%)	90% CI (%)
	Ainuovirine Alone Period 1 N = 18	Ainuovirine + Fluconazole Period 2 N = 17			Fluconazole Alone Period 1 (Group B) N = 18	Fluconazole + Ainuovirine Period 2 (Group B) N = 18		
$C_{max,ss}$ (mg/L)	0.70	1.63	233.0	203.3–267.1	12.59	12.53	99.5	96.4–102.7
$AUC_{0-24,ss}$ (h mg/L)	7.46	26.08	349.6	293.0–417.2	211.93	215.87	101.9	99.5–104.3
$T_{max,ss}$ (h)	2.0 (0.5, 4.0)	3.0 (1.0, 4.0)	NA	NA	1.0 (0.5, 3.0)	1.5 (0.5, 4.0)	NA	NA

Notes: Data are GM (geometric mean) except for T_{max} where median (minimum, maximum) values are presented.

Abbreviations: CI, confidence interval; PK, pharmacokinetic; $C_{max,ss}$, maximum steady-state plasma concentration; $AUC_{0-24,ss}$, area under the plasma concentration-time curve from time 0 to 24 h; $T_{max,ss}$, time to reach maximum concentration at steady-state.

Table 4 Summary of TEAE in Group A (Safety Analysis Set)

SOC/PT	Ainuovirine 150 mg Period 1 N= 18, n (%)	Ainuovirine 150 mg Plus Fluconazole 200 mg Period 2 N= 18, n (%)	Total N= 18, n (%)
Subjects with TEAE	11 (61.1)	15 (83.3)	15 (83.3)
Subjects with ainuovirine-related TEAE	11 (61.1)	15 (83.3)	15 (83.3)
Subjects with fluconazole-related TEAE	-	14 (77.8)	14 (77.8)
TEAE of CTCAE grade 3 or more	0	0	0
Ainuovirine-related adverse events of CTCAE grade 3 or more	0	0	0
Fluconazole-related adverse events of CTCAE grade 3 or more	-	0	0
SAE	0	0	0
Ainuovirine-related SAE	0	0	0
Fluconazole-related SAE	-	0	0
Concomitant medications, non-pharmacological treatment related TEAE	0	2 (11.1)	2 (11.1)

Notes: n number of subjects with an adverse event, % proportion of subjects having an adverse event.

Abbreviations: TEAE, treatment-emergent adverse events; SOC, system organ class; PT, preferred term; CTCAE, Common Terminology Criteria for AEs; SAE, serious adverse event.

Table 5 Summary of TEAE in Group B (Safety Analysis Set)

SOC/PT	Fluconazole 200 mg Period 1 N= 18, n (%)	Fluconazole 200 mg Plus Ainauvirine 150 mg Period 2 N= 18, n (%)	Total N= 18, n (%)
Subjects with TEAE	10 (55.6)	15 (83.3)	17 (94.4)
Subjects with ainauvirine-related TEAE	-	12 (66.7)	12 (66.7)
Subjects with fluconazole-related TEAE	9 (50.0)	15 (83.3)	16 (88.9)
TEAE of CTCAE grade 3 or more	0	0	0
Ainauvirine-related adverse events of CTCAE grade 3 or more	-	0	0
Fluconazole-related adverse events of CTCAE grade 3 or more	0	0	0
SAE	0	0	0
Ainauvirine-related SAE	-	0	0
Fluconazole-related SAE	0	0	0
Concomitant medications, non-pharmacological treatment related TEAE	0	2 (11.1)	2 (11.1)

Notes: n number of subjects with an adverse event, % proportion of subjects having an adverse event.

Abbreviations: TEAE, treatment-emergent adverse events; SOC, system organ class; PT, preferred term; CTCAE, Common Terminology Criteria for AEs; SAE, serious adverse event.

PBPK Model Development and Evaluation

The input parameters for ainauvirine PBPK model are summarized in [Tables S3](#) and [S4](#). The final PBPK model of ainauvirine has been developed to predict concentration–time profiles following oral administration of 75 mg and 150 mg of ainauvirine in the supplementary material ([Figure S1a–f](#)). Predictions of plasma concentration–time trajectories for different doses were in close agreement with the observed data. The observed plasma concentration–time datasets fell almost within the 90% prediction intervals (5th and 95th percentiles) for ainauvirine concentrations predicted by the model. The ratios of predicted and observed value of $C_{\max}$ and AUC were all within 0.5- to 2-fold ([Table S5](#)).

These simulations demonstrated that the PBPK model's ability to characterize dose-dependent PK changes of ainauvirine and describe its plasma exposure in clinical studies for both single and multiple doses. Subsequently, a previously reported PBPK model of fluconazole was integrated with the model of ainauvirine to assess their potential DDI ([Figure S1g](#)). The input parameters for fluconazole PBPK model are summarized in [Table S6](#). The observed individual plasma concentration–time datasets almost fell entirely within the 90% prediction intervals (5th and 95th percentiles) for ainauvirine concentrations predicted by the model ([Figure S1g](#)). The presented PBPK model can capture the peak, ie., better describe $T_{\max}$ and $C_{\max}$ of ainauvirine PK. The ratio of the predicted value (1.66 mg/L) to the observed value (1.67 mg/L) for $C_{\max,ss}$ in group A (Period 2) was 1.00. Moreover, the ratio of the predicted value (29.69 h·mg/L) to the observed value (26.7 h·mg/L) for $AUC_{0-\infty,ss}$ in group A (Period 2) was 1.11. The ratios of predicted and observed value of $C_{\max,ss}$ and $AUC_{0-\infty,ss}$ were all within 0.5- to 2-fold. These simulations indicated that the model could capture ainauvirine's PK characteristics involving plasma exposure and its variability, as well as DDI induced by fluconazole's inhibition of CYP2C19.

The Ainauvirine and Fluconazole DDI

Dose reduction of ainauvirine when it was co-administered with fluconazole was considered. Two clinical practice scenarios were simulated using the final PBPK model. Scenario 1 focused on a PLWH with fungal infection who was on long-term ainauvirine therapy and needed to initiate fluconazole. Scenario 2 focused on a treatment-naïve PLWH with a history of fungal infection who received fluconazole prior to initiating ainauvirine.

At steady state of fluconazole (200 mg q.d.), the dose of ainauvirine should be reduced to decrease its exposure. The final PBPK model of ainauvirine was applied to predict DDI with fluconazole following the administration of lower multiple doses of 20 mg q.d. of ainauvirine. In scenario 1, subjects received ainauvirine 150 mg q.d. for 7 days, and then received fluconazole (200 mg q.d.) concomitantly with ainauvirine (20 mg q.d.) from day 8 to day 22. In DDI scenario 1, the predicted $AUC_{0-\tau,ss}$ (7.74 h·mg/L) and $C_{\max,ss}$ (0.38 mg/L) of ainauvirine were significantly reduced when the adjusted dose of 20 mg was applied ([Figure 3](#) and [Table 6](#)), compared to those observed with the 150 mg dose in the current study. Moreover, the predicted C_{trough} (0.28 mg/L) of ainauvirine was higher than that observed (0.19 mg/L) in

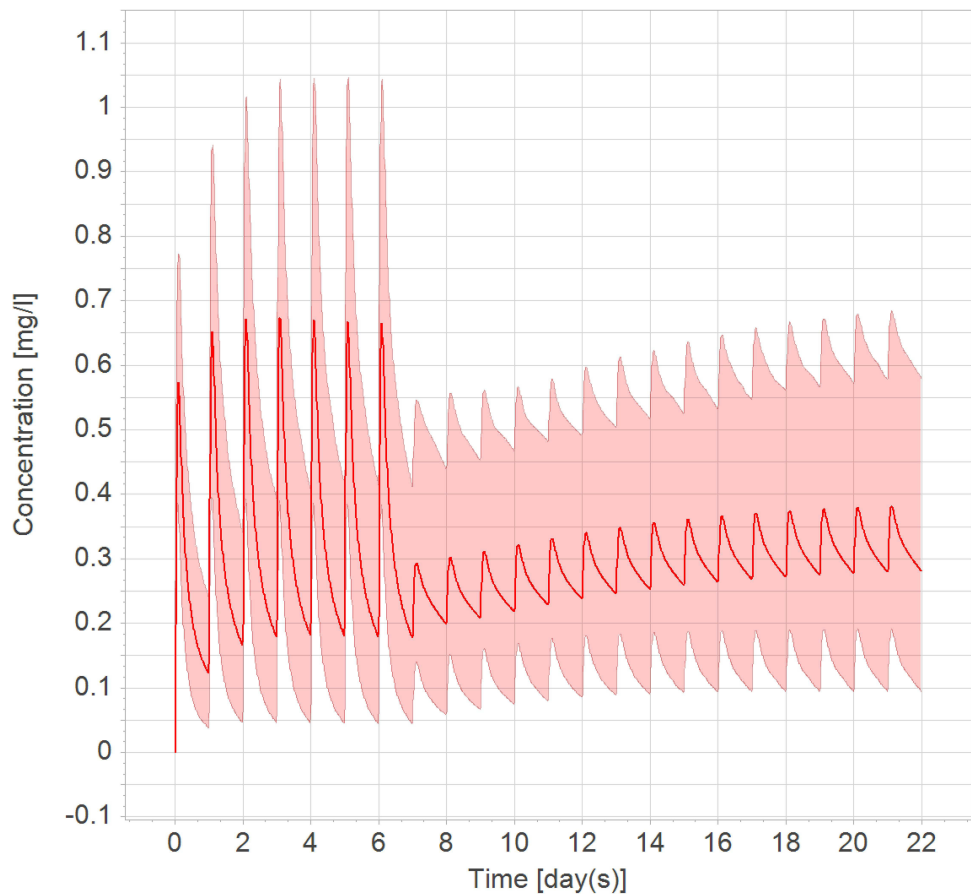


Figure 3 Predicted concentration-time profiles following multiple doses of ainuovirine in the absence/presence of coadministered fluconazole were simulated for virtual DDI scenario 1 (i.e., ainuovirine po 150 mg q.d. Day 1–7; ainuovirine po 20 mg q.d. + fluconazole 200 mg q.d. Day 8–22). Red solid line and red shaded area represent the arithmetic mean and the 90% prediction interval (5th and 95th percentiles) of the model based simulated ainuovirine concentrations ($n = 1000$), respectively.

the current study, indicating that the therapeutic drug level was maintained. In the previous Phase 1 study of ainuovirine conducted in China (ADYY-ACC007-103, unpublsh), when treatment-naïve HIV-1-positive adults were treated with a 150 mg dose of ainuovirine on day 10, the $C_{max,ss}$ and $AUC_{0-\tau,ss}$ of ainuovirine were 0.53 ± 0.14 mg/L and 6.29 ± 2.42 h·mg/L, respectively (Table 6).

In scenario 2, subjects received fluconazole 200 mg q.d. for 7 days and then received fluconazole (200 mg q.d.) concomitantly with ainuovirine (20 mg q.d.) from day 8 to day 22 (Figure 4). The simulated plasma concentrations of ainuovirine over time reached steady state on day 21 with the $C_{max,ss}$ of 0.36 mg/L and the $AUC_{0-\tau,ss}$ of 7.28 h·mg/L

Table 6 Comparison of Mean Predicted (Scenario 1) Vs. Observed $C_{max,ss}$, $AUC_{0-\tau,ss}$ and C_{trough} of Ainuovirine in the Absence and Presence of Coadministration of Fluconazole

Ainuovirine	Number of Subjects	$C_{max,ss}$			$AUC_{0-\tau,ss}$			C_{trough}		
		Pred (mg/L)	Obs (mg/L)	$R_{pred/obs}$	Pred (h mg/L)	Obs (h mg/L)	$R_{pred/obs}$	Pred (mg/L)	Obs (mg/L)	$R_{pred/obs}$
Multiple oral doses 150 mg q.d. from day 1–7	18	0.38	0.73	0.52	7.74	8.02	0.97	0.28	0.19	1.47
Multiple oral doses 150 mg q.d. from day 1–10 (ADYY-ACC007-103, unpublsh)	10	0.38	0.53	0.73	7.74	6.29	1.23	0.28	0.15	1.82

Abbreviations: q.d, once daily; $C_{max,ss}$, maximum steady-state plasma concentration; $AUC_{0-\tau,ss}$, area under the plasma concentration-time curve from time 0 to τ (dosing interval) at steady state; C_{trough} , plasma trough concentration; Pred, prediction; Obs, observation; $R_{pred/obs}$, the ratio of the prediction value and the observation value.

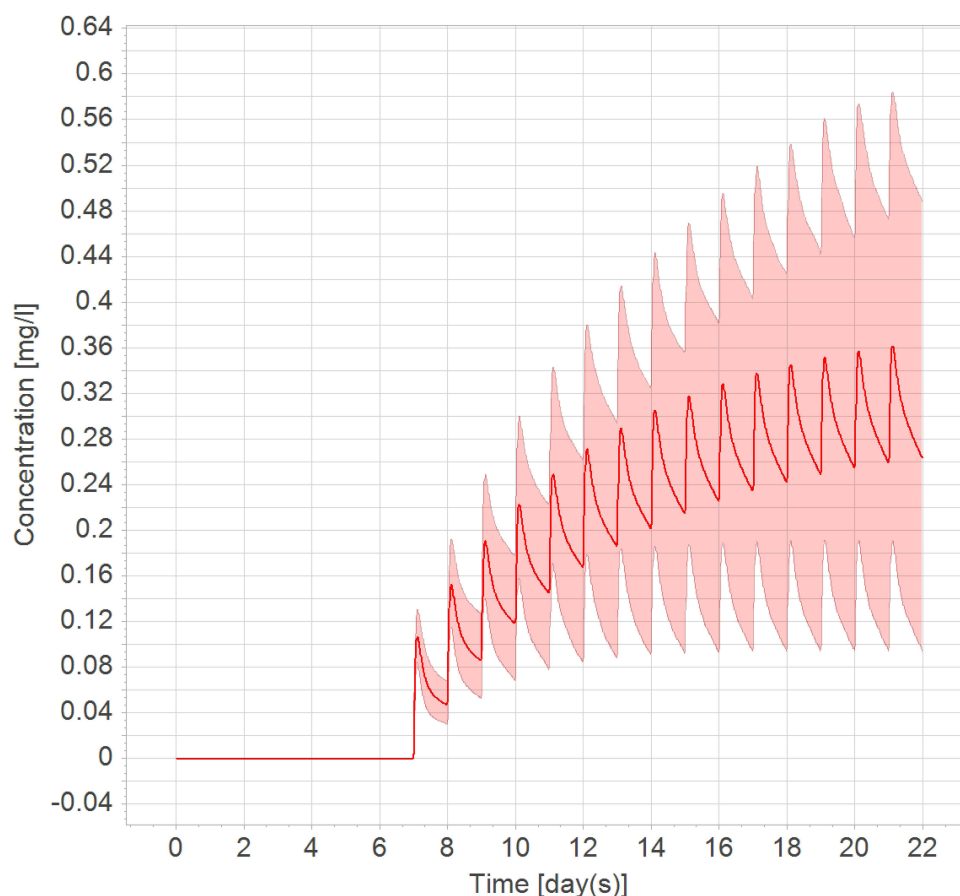


Figure 4 Ainuovirine concentration-time profiles were simulated for virtual DDI scenario 2 (i.e., fluconazole 200 mg Day 1–7; fluconazole 200 mg q.d. + ainuovirine po 20 mg q.d. Day 8–22). Red solid line and red shaded area represent the arithmetic mean and the 90% prediction interval (5th and 95th percentiles) of the model based simulated ainuovirine concentrations ($n = 1000$), respectively.

(Figure 4 and Table 7). Moreover, the predicted ainuovirine C_{trough} (0.26 mg/L) was higher than that observed (0.19 mg/L) in the current study, indicating that the therapeutic drug level was maintained.

The predicted values of ainuovirine (20 mg q.d.) $AUC_{0-\text{tau},ss}$ and $C_{\text{max},ss}$ when coadministered with 200 mg fluconazole in both scenarios were similar (Tables 6 and 7). The ratios of predicted to observed $C_{\text{max},ss}$, $AUC_{0-\text{tau},ss}$ and C_{trough} in group A and in the previous phase 1 study of ainuovirine (ADYY-ACC007-103, unpublished) were all within the 0.5- to 2.0-fold limits (Tables 6 and 7).

Table 7 Comparison of Mean Predicted (Scenario 2) vs. Observed $C_{\text{max},ss}$, $AUC_{0-\text{tau},ss}$ and C_{trough} of Ainuovirine in the Absence and Presence of Coadministration of Fluconazole

Ainuovirine	Number of Subjects	$C_{\text{max},ss}$			$AUC_{0-\text{tau},ss}$			C_{trough}		
		Pred (mg/L)	Obs (mg/L)	$R_{\text{pred/obs}}$	Pred (h mg/L)	Obs (h mg/L)	$R_{\text{pred/obs}}$	Pred (mg/L)	Obs (mg/L)	$R_{\text{pred/obs}}$
Multiple oral doses 150 mg q.d. from day 1–7	18	0.36	0.73	0.50	7.28	8.02	0.91	0.26	0.19	1.36
Multiple oral doses 150 mg q.d. from day 1–10 (ADYY-ACC007-103, unpublished)	10	0.36	0.53	0.69	7.28	6.29	1.16	0.26	0.15	1.69

Abbreviations: q.d, once daily; $C_{\text{max},ss}$, maximum steady-state plasma concentration; $AUC_{0-\text{tau},ss}$, area under the plasma concentration-time curve from time 0 to tau (dosing interval) at steady state; C_{trough} , plasma trough concentration; Pred, prediction; Obs, observation; $R_{\text{pred/obs}}$, the ratio of the prediction value and the observation value.

Discussion

Ainuovirine is primarily metabolized by CYP2C19.³¹ It is well-tolerated over a wide concentration range (dose range: 75–600 mg) in healthy subjects.¹³ Fluconazole is a strong CYP2C19 inhibitor.¹⁸ In the present clinical study, the geometric mean $AUC_{0-24,ss}$ of ainuovirine (150 mg q.d.) was increased 3.5-fold in the presence of fluconazole (200 mg q.d.). The incidence of TEAEs was higher when ainuovirine was coadministered with fluconazole compared to when ainuovirine was given alone (Tables 4 and S1). These findings demonstrated a significant PK interaction with ainuovirine when fluconazole was administered to healthy adults and this increase in ainuovirine exposure was deemed to have safety consequences. Therefore, dose reduction of ainuovirine is warranted when coadministered with fluconazole.

After coadministration with fluconazole, the value of CL_{ss}/F for ainuovirine decreased from 21.66 ± 8.71 to 5.73 ± 0.81 L/h, and the value of Vz_{ss}/F for ainuovirine decreased from 518.56 ± 246.57 to 227.12 ± 107.73 L. The approximately 74% reduction in CL_{ss}/F demonstrated decreased systemic clearance and slower elimination during fluconazole co-administration. The decrease in Vz_{ss}/F indicated an increase in the oral bioavailability (F) caused by decreased pre-systemic metabolism. The value of $t_{1/2,ss}$ for ainuovirine increased from 17.3 ± 6.2 to 27.3 ± 10.7 h, however, the extent of this prolongation (~1.6-fold) is less than the increase in $AUC_{0-24,ss}$ (~3.3-fold). This suggests that the interaction is not solely due to inhibition of elimination but also involves significant inhibition of first-pass metabolism (both hepatic and intestinal), which is consistent with the observed decrease in CL_{ss}/F and Vz_{ss}/F . The increase in ainuovirine exposure when coadministered with fluconazole is predominantly due to the strong inhibition of CYP2C19, leading to reduced metabolism and clearance. These results demonstrated that fluconazole has a significant effect on the steady-state PK of ainuovirine. Clinically significant drug interactions have been reported when fluconazole is coadministered with other drugs that are substrates for CYP2C19, such as nelfinavir²¹ and omeprazole.²⁰ Coadministration with fluconazole (100 mg q.d.) significantly increased omeprazole C_{max} by 240% and AUC by 629%, with a threefold prolongation of $t_{1/2}$.²⁰ However, the detailed information regarding the CYP2C19 genotypes of the subjects was not included in their studies.

It has been reported that ainuovirine is slowly metabolized by CYP2C19; genetic polymorphisms in this enzyme have relatively little effect on its clearance kinetics.¹¹ However, in the present clinical study, the interindividual variability (coefficient of variation, %CV) for $C_{min,ss}$, $C_{max,ss}$, and $AUC_{0-24,ss}$ was 52%, 29%, and 39% for ainuovirine alone, and 15%, 18%, and 15% for the combination of ainuovirine plus fluconazole (group A in Table 2), indicated reduced pharmacokinetic variability during the combination therapy. The interindividual variability in ainuovirine PK parameters was higher when administered alone than during co-administration with fluconazole. Fluconazole as a potent CYP2C19 inhibitor may cause phenoconversion.^{32,33} When 200 mg of fluconazole was coadministered, the AUC of flurbiprofen in predicted CYP2C9 normal metabolizers ($n = 11$) increased to a level comparable to that observed in CYP2C9 intermediate metabolizers ($n = 8$). Moreover, a 400 mg dose of fluconazole further elevated the AUC, bringing it closer to the levels seen in CYP2C9 poor metabolizers ($n = 2$).³⁴ The higher interindividual variability in ainuovirine PK parameters when administered alone vs. with fluconazole coadministration may be primarily attributed to the phenoconversion induced by fluconazole coadministration. By inhibiting CYP2C19 activity, fluconazole (200 mg q.d.) masks genetic differences, thereby reducing variability in ainuovirine exposure regardless of CYP2C19 genotype. As coadministration with fluconazole did not increase variability, the effect of CYP2C19 polymorphisms on the pharmacokinetics of ainuovirine in the DDI study with fluconazole could be small. In addition, in Group A of the present study, PK analysis was also conducted in 12 subjects from Han ethnicity and 6 subjects from other ethnicities. No significant differences were observed between Han ethnicity ($n=12$) and other ethnicities ($n=6$) in $C_{min,ss}$, $C_{max,ss}$, and $AUC_{0-24,ss}$ whether for ainuovirine alone on Day 7 or for the combination of ainuovirine and fluconazole on Day 14 (data not shown). Thus, the effect of ethnic groups on the PK of ainuovirine in the DDI study with fluconazole was small in the present study. However, the sample size of other ethnic groups in this trial was very limited; thus, the results may not be fully representative of the general ethnic population.

This study also evaluated the influence of ainuovirine on the PK of fluconazole. Results demonstrated that the effect of ainuovirine on the pharmacokinetics of fluconazole was minimal. It could be due to the fact that fluconazole is

predominantly excreted renally, with only 11% of the drug being metabolized by the liver.³⁵ In addition, the combination of ainoovirine and fluconazole was generally well-tolerated for 14 days without any SAEs in this study.

Given the increased risk of fungal infections, PLWH usually require concomitant antifungal therapy or prophylaxis. Current therapies for candidiasis and cryptococcal diseases mostly use fluconazole.^{14,15} Clinically significant drug interactions have also been reported when fluconazole is coadministered with other NNRTIs. For example, the coadministration of fluconazole (200 mg q.d.) significantly increased the $C_{\max}$ and AUC_{12h} of etravirine 1.75-fold and 1.86-fold, respectively,³⁶ and increased the C_{trough} of nevirapine 1.51-fold.³⁷ The coadministration of fluconazole increased the AUC of efavirenz 1.16-fold.³⁸ Among them, the increase in etravirine exposure in the presence of fluconazole did not have a clinically meaningful effect.³⁶ However, the increases in $C_{\max}$ and AUC_{0-24} of ainoovirine in this study are higher than those observed for other NNRTIs in similar Phase I studies. Dose reduction of ainoovirine is necessary when coadministered with the antifungal drug fluconazole.

A PBPK model of ainoovirine integrating the model of fluconazole was developed and population PK analysis was performed. The model was used to predict plasma concentrations of ainoovirine and DDI between ainoovirine and fluconazole. A criterion that predicted/observed AUC and $C_{\max,ss}$ ratios are within a twofold (0.5–2.0) range is widely accepted in the field of PBPK modelling.^{39,40} This criterion was used in the current study. More importantly, for antiretroviral drugs, C_{trough} is critical for the maintenance of HIV-1 suppression; thus, the predicted/observed C_{trough} ratio should be considered in the present study. In the present study, the PK profiles fit with the model (Figure S1) and the PBPK model estimates the mean $AUC_{0-\tau,ss}$, $C_{\max,ss}$, and C_{trough} ratios of ainoovirine with acceptable precision. The predictive performance of the final PBPK model for the interaction scenario is excellent, because the ratios of predicted-to-observed $C_{\max,ss}$ and $AUC_{0-\infty,ss}$ are very close to 1.0 (1.00 and 1.11, respectively), which provides high confidence in the model's simulations for dose optimization. The clinical observation of ainoovirine is based on its Phase 1 study (ADYY-ACC007-103) and the current clinical study. The mean predicted/observed $AUC_{0-\tau,ss}$, $C_{\max,ss}$, and C_{trough} ratios are within 0.5- to 2.0-fold in Tables 6 and 7, indicating the model can simulate plasma concentrations of ainoovirine and predict DDI with fluconazole. Based on the findings of PBPK modelling, the model-predicted C_{trough} for the 20 mg dose under fluconazole inhibition is higher than the observed C_{trough} for the 150 mg dose alone, indicating that the reduced dose will maintain efficacy while preventing overexposure. Although the presented DDI model can predict the exposure of ainoovirine when coadministered with fluconazole, further clinical studies are needed to verify the model and to confirm the applicability of its predictions for individual patients. The major limitation of this study is that the subjects in this trial were predominantly Chinese healthy individuals and it would have been beneficial if the determination of CYP2C19 genotypes of the enrolled healthy individuals had been included.^{41,42} An analysis of the effect of different genotypes on the degree of DDI would have been of interest.

Data Sharing Statement

The datasets in this study are available from the corresponding author upon reasonable request.

Ethics Approval and Informed Consent

This study was registered at chinactr.org.cn (Registration number: ChiCTR2300070530). The study protocol and subject consent forms were reviewed and approved by the Medical Ethics Committees of Guangzhou Eighth People's Hospital (ethical approval batch number: 202302196), Guangzhou Medical University, and conducted in accordance with the Declaration of Helsinki and the International Conference on Harmonization (ICH) Good Clinical Practice Guidelines (GCP) and local regulatory requirements. All subjects provided written informed consent form.

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Author Contributions

All authors made a significant contribution to the work reported, whether that is in the conception, study design, execution, acquisition of data, analysis and interpretation, or in all these areas; took part in drafting, revising or critically

reviewing the article; gave final approval of the version to be published; have agreed on the journal to which the article has been submitted; and agree to be accountable for all aspects of the work.

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Disclosure

The authors report no conflicts of interest.

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